1) $V^{t+\frac{\Delta t}{2}} = V^{t-\frac{\Delta t}{2}} + \frac{\Delta t}{M} \cdot R^{t}$

$$\boxed{M = \frac{\Delta t^{2}}{2} K}$$

2) $X^{t+\Delta t} = X^{t} + \Delta t \cdot V^{t+\frac{\Delta t}{2}}$

$$\boxed{V^{t-\frac{\Delta t}{2}} + \frac{2}{\Delta t} K^{-1} \cdot R^{t}}$$

$$\frac{1}{\Delta t} K^{-1}(1+A)$$

$$A = \frac{1-\frac{c}{2}}{1+\frac{c}{2}}$$

when $c=0$ $A=1$

$$V^{t+\frac{\Delta t}{2}} = A V^{t-\frac{\Delta t}{2}} + B \cdot R^{t}$$

$$B = \frac{\Delta t}{2M}(1+A)$$

$$B = \frac{\Delta t}{M}$$

$$x^{t+\Delta t} = x^{t} + \Delta t \cdot A \cdot V^{t-\frac{\Delta t}{2}} + K^{-1}(1+A) R^{t}$$

$$X^{t+\Delta t} = x^{t} + \Delta t \cdot V^{t-\frac{\Delta t}{2}} + \boxed{2K^{-1} \cdot R^{t}}$$

$t=0$

1) $V^{t+\frac{\Delta t}{2}} = V^{t-\frac{\Delta t}{2}} + \frac{\Delta t}{M} \cdot R^{t}$

if $V^{\frac{\Delta t}{2}} = -V^{-\frac{\Delta t}{2}}$ $\quad 2V^{\frac{\Delta t}{2}} = \frac{\Delta t}{M} R^{t}$

$$V^{\frac{\Delta t}{2}} = \frac{\Delta t}{2M} \cdot R^{t} = \frac{1}{\Delta t} \cdot K^{-1} \cdot R^{t}$$

$$X^{\Delta t} = X^{t} + \Delta t \cdot V^{\frac{\Delta t}{2}} = \boxed{X^{t} + K^{-1} R^{t}}$$

Start